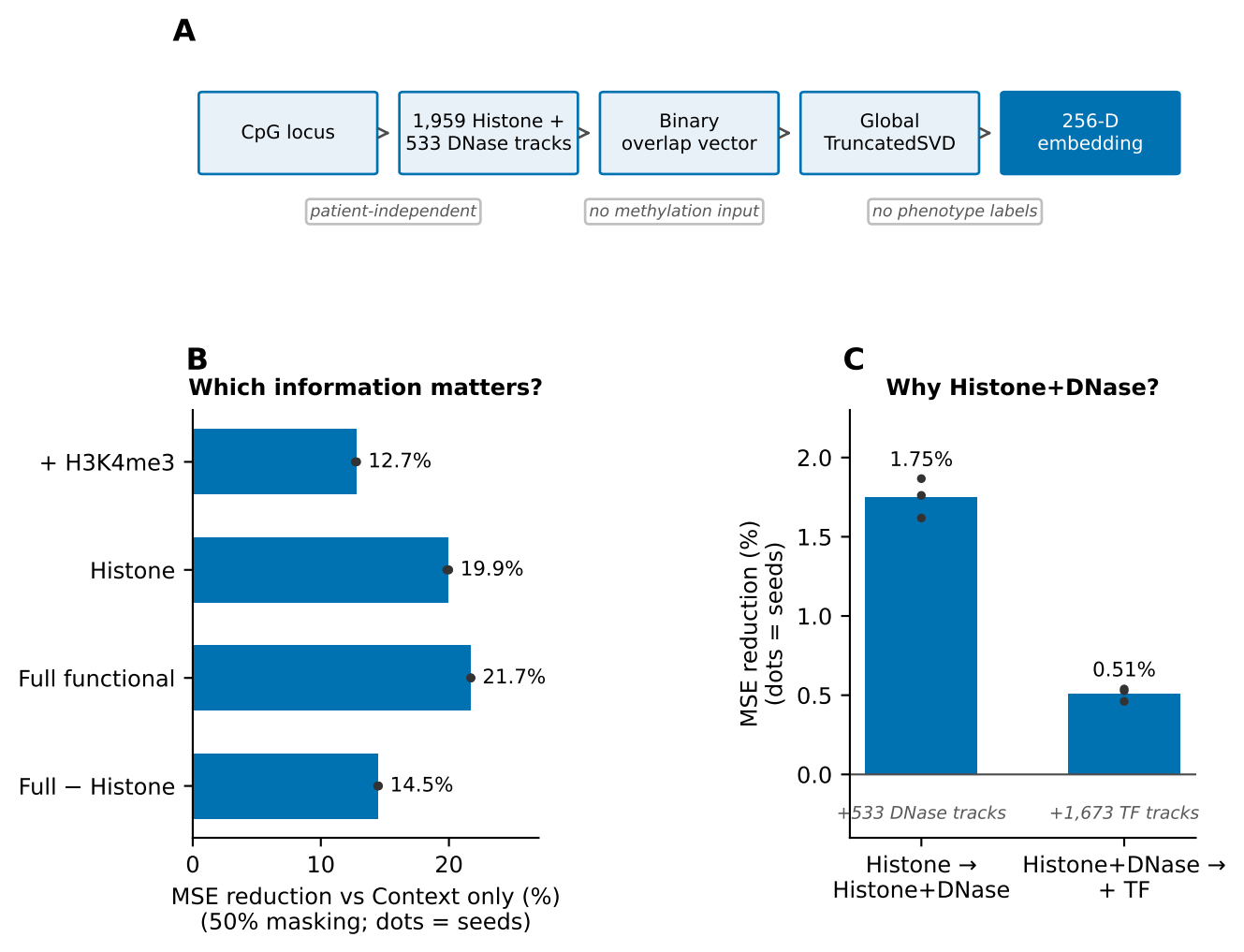
A
CpG locus
1,959 Histone + 533 DNase tracks
Binary overlap vector
Global TruncatedSVD
256-D embedding
patient-independent
no methylation input
no phenotype labels
B
Which information matters?
+ H3K4me3
Histone
Full functional
Full − Histone
12.7%
19.9%
21.7%
14.5%
0
10
20
MSE reduction vs Context only (%)
(50% masking; dots = seeds)
C
Why Histone+DNase?
MSE reduction (%)
(dots = seeds)
2.0
1.5
1.0
0.5
0.0
1.75%
0.51%
+533 DNase tracks
+1,673 TF tracks
Histone → Histone+DNase
Histone+DNase → + TF